

K Theory

Book Notes of *K-Theory*

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1 Vector Bundles

1.1 Basic Definition

Let $E = X \times V$, $F = X \times W$ be two trivial vector bundles over a space X , then any homomorphism $\varphi : E \rightarrow F$ determines a continuous map

$$\Phi : X \rightarrow \text{Hom}(V, W)$$

by the formula

$$\varphi(x, v) = (x, \Phi(x)v).$$

Conversely, any continuous map $\Phi : X \rightarrow \text{Hom}(V, W)$ defines a homomorphism $\Phi : E \rightarrow F$.

Let $\text{Iso}(V, W) \subset \text{Hom}(V, W)$ be the set of all isomorphisms from V to W . Then $\text{Iso}(V, W)$ is an open subset of $\text{Hom}(V, W)$. Thus for any homomorphism φ , the set of points $x \in X$ such that $\varphi_x : E_x \rightarrow F_x$ is an isomorphism form an open subset of X .

All of these assertions are local in nature, and therefore are valid for arbitrary vector bundles.

1.2 Operations on vector bundles

Sections of $\text{Hom}(E, F)$ correspond in a one-to-one fashion with homomorphisms $\varphi : E \rightarrow F$. We define $\text{HOM}(E, F)$ to be the vector space of all homomorphisms from E to F . Notice that $\text{HOM}(E, F) = \Gamma(\text{Hom}(E, F))$.

1.3 Sub-bundles and quotient bundles

Let $\varphi : E \rightarrow F$ be arbitrary homomorphism, the function $x \mapsto \dim(\ker(\varphi_x))$ need not be constant, or even locally constant.

In fact, $\dim(\ker(\varphi_x))$ is a lower semi-continuous function of x , while $\dim(\text{im}(\varphi_x))$ is upper semi-continuous.

$$\forall x \in X, \exists U \ni x, \text{ s.t. } \forall y \in U, \dim(\text{im}(\varphi_x)) \leq \dim(\text{im}(\varphi_y))$$

Definition 1.1. We say $\varphi : E \rightarrow F$ is a strict homomorphism if $\dim(\ker(\varphi_x))$ is locally constant.

Proposition 1.2. If $\varphi : E \rightarrow F$ is strict, then

- $\ker(\varphi)$ is a sub-bundle of E ,
- $\text{im}(\varphi)$ is a sub-bundle of F ,
- $\text{coker}(\varphi) = F/\text{im}(\varphi)$ is a quotient bundle of F .

1.4 Vector Bundles on compact spaces

From now on, we shall assume that all spaces are compact Hausdorff spaces.

Lemma 1.3. *Let Y be a compact Hausdorff space, $f_t : Y \rightarrow X$ ($0 \leq t \leq 1$) be a homotopy, and let E be a vector bundle over X . Then*

$$f_0 * E \cong f_1 * E.$$

We can construct vector bundles by gluing together trivial bundles or clutching. Let

$$X = X_1 \cup X_2, \quad A = X_1 \cap X_2,$$

all of which are compact Hausdorff spaces. Let E_1, E_2 be vector bundles over X_1, X_2 respectively, and let $\varphi : E_1|_A \rightarrow E_2|_A$ be an isomorphism. Then we can construct $E_1 \cup_\varphi E_2$ over X .

Proposition 1.4. • *If $\beta_i : E_i \rightarrow E'_i$ are isomorphisms ($i = 1, 2$) such that $\beta_2|_A \circ \varphi = \varphi' \circ \beta_1|_A$, then*

$$E_1 \cup_\varphi E_2 \cong E'_1 \cup_{\varphi'} E'_2$$

- *If (E_i, φ) and (E'_i, φ') are two clutching data over the same X_i and A , then*

$$\begin{aligned} (E_1 \cup_\varphi E_2) \oplus (E'_1 \cup_{\varphi'} E'_2) &\cong (E_1 \oplus E'_1) \cup_{\varphi \oplus \varphi'} (E_2 \oplus E'_2) \\ (E_1 \cup_\varphi E_2) \otimes (E'_1 \cup_{\varphi'} E'_2) &\cong (E_1 \otimes E'_1) \cup_{\varphi \otimes \varphi'} (E_2 \otimes E'_2) \\ (E_1 \cup_\varphi E_2)^* &\cong E_1^* \cup_{(\varphi^*)^{-1}} E_2^* \end{aligned}$$

Lemma 1.5. *The isomorphism class of $E_1 \cup_\varphi E_2$ depends only on the homotopy class of $\varphi : E_1|_A \rightarrow E_2|_A$.*

Lemma 1.6. *For any space X , there is a natural isomorphism*

$$\text{Vect}_n(S(X)) \cong [X, GL_n(\mathbb{C})],$$

Definition 1.7. A subspace $V \subset \Gamma(E)$ is said to be ample if

$$\varphi : X \times V \rightarrow E, \quad (x, s) \mapsto s(x)$$

is surjective.

Lemma 1.8. *For any vector bundle E over a compact Hausdorff space X , there exists a finite-dimensional ample subspace $V \subset \Gamma(E)$.*

Proof. Finite local triviality of E , and partition of unity on X . □

Corollary 1.9. *For any vector bundle E , there exists an epimorphism $\varphi : X \times \mathbb{C}^m \rightarrow E$ for some m .*

Corollary 1.10. *For any vector bundle E , there exists a vector bundle F such that $E \oplus F$ is trivial.*